

PRACTICAL 8

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1. Develop a program using malloc(), calloc(), realloc(), and free(). Monitor memory allocation behavior and identify memory leaks using Valgrind.

Develop a program that demonstrates Copy-on-Write behavior after a fork(). Analyze memory usage before and after modifying data in the child process.

```
GNU
if (a == NULL) return 1;
printf("malloc: 5 integers allocated\n");

for (int i = 0; i < 5; i++)
    a[i] = i + 1;

b = calloc(5, sizeof(int));
if (b == NULL) {
    free(a);
    return 1;
}
printf("calloc: 5 integers allocated and initialized to zero\n");

a = realloc(a, 10 * sizeof(int));
if (a == NULL) {
    free(b);
    return 1;
}
printf("realloc: memory increased from 5 to 10 integers\n");

for (int i = 5; i < 10; i++)
    a[i] = i + 1;

temp = malloc(100 * sizeof(int));
printf("malloc: extra memory allocated\n");

free(a);
printf("free: first memory block released\n");
```

```

    free(b);
    printf("free: second memory block released\n");

    printf("Program completed.\n");

    return 0;
}

```

```

sathvik@DESKTOP-8GJTBQP: ~$
sathvik@DESKTOP-8GJTBQP: ~$
sathvik@DESKTOP-8GJTBQP: ~$
sathvik@DESKTOP-8GJTBQP: ~$ cd OSLab
sathvik@DESKTOP-8GJTBQP: ~/OSLab$ cd Practical
sathvik@DESKTOP-8GJTBQP: ~/OSLab/Practical$ nano Problem8
sathvik@DESKTOP-8GJTBQP: ~/OSLab/Practical$ gcc Problem8.c -o Problem8
sathvik@DESKTOP-8GJTBQP: ~/OSLab/Practical$ ./Problem8
-Bash: ./Problem8.c: Permission denied
sathvik@DESKTOP-8GJTBQP: ~/OSLab/Practical$ gcc Problem8.c -o Problem8
./Problem8
sathvik@DESKTOP-8GJTBQP: ~/OSLab/Practical$ nano Problem8.c
sathvik@DESKTOP-8GJTBQP: ~/OSLab/Practical$ gcc Problem8.c -o Problem8
sathvik@DESKTOP-8GJTBQP: ~/OSLab/Practical$ ./Problem8
Starting memory allocation...
malloc: 5 integers allocated
calloc: 5 integers allocated and initialized to zero
realloc: memory increased from 5 to 10 integers
malloc: extra memory allocated
free: first memory block released
free: second memory block released
Program completed.
sathvik@DESKTOP-8GJTBQP: ~/OSLab/Practical$ nano Problem8.c
sathvik@DESKTOP-8GJTBQP: ~/OSLab/Practical$

```

8b) #include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/wait.h>

#define SIZE 1000000

```
int main() {  
    int *arr = malloc(SIZE * sizeof(int));  
  
    if (arr == NULL) {  
        perror("malloc");  
        return 1;  
    }  
  
    for (int i = 0; i < SIZE; i++)  
        arr[i] = i;  
  
    printf("=== Parent Process ===\n");  
    printf("Parent PID: %d\n", getpid());  
    printf("Memory allocated: %lu bytes\n",  
        (unsigned long)(SIZE * sizeof(int)));  
    printf("Initial arr[0] = %d\n", arr[0]);  
  
    fflush(stdout);
```

```
pid_t pid = fork();
```

```
if (pid < 0) {  
    perror("fork");  
    free(arr);  
    return 1;  
}
```

```
if (pid == 0) {  
    printf("\n=== Child Process ===\n");  
    printf("Child PID: %d\n", getpid());  
    printf("Before modification: arr[0] = %d\n", arr[0]);
```

```
    arr[0] = 9999;
```

```
    printf("After modification: arr[0] = %d\n", arr[0]);  
    printf("Child has its own copy of the modified  
memory page.\n");
```

```
    fflush(stdout);
    free(arr);
    exit(0);
}
else {
    wait(NULL);

    printf("\n=== Parent Process After Child ===\n");
    printf("Parent PID: %d\n", getpid());
    printf("Parent arr[0] = %d\n", arr[0]);
    printf("Parent data remains unchanged.\n");

    fflush(stdout);
    free(arr);
}

return 0;
}
```

```
sathvik@DESKTOP-8GJTBQP:~/OSLab$ cd Practical
sathvik@DESKTOP-8GJTBQP:~/OSLab/Practical$ nano Practical8b.c
sathvik@DESKTOP-8GJTBQP:~/OSLab/Practical$ gcc Practical8b.c
sathvik@DESKTOP-8GJTBQP:~/OSLab/Practical$ gcc Practical8b.c -o Practical8b
sathvik@DESKTOP-8GJTBQP:~/OSLab/Practical$ ./Practical8b

=== Parent Process ===
Parent PID: 532
Memory allocated: 4000000 bytes
Initial arr[0] = 0

=== Child Process ===
Child PID: 533
Before modification: arr[0] = 0
After modification: arr[0] = 9999
Child has its own copy of the modified memory page.

=== Parent Process After Child ===
Parent PID: 532
Parent arr[0] = 0
Parent data remains unchanged.
sathvik@DESKTOP-8GJTBQP:~/OSLab/Practical$ |
```